Residual Exposure & Spot Inventory

Change report for `tests/test_residual_exposure.py` and `tests/test_spot_inventory.py` · crypto_MFT_arb_platform · 12 Oct 2026
GitHub: vaibhaw1994kumar/CryptoMFT · commits `40b52d4`, `fa70b81`, `8d66ce7` · evidence bundle `reports/inventory_residual_evidence.zip`

17 / 17	250 / 250	≤ 0.001 BTC	±0.2 USDT
new tests passing (residual 7 + inventory 10)	crypto_stat_arb suite; offline acceptance gate PASSED	end residual on replayed cycles (was up to 0.007 vs 0.002 cap)	plan net_pnl vs equity PnL (was up to 551 USDT apart)

1. Summary

Two capital/risk-layer gaps were fixed. **(1) Residual exposure:** a balanced two-leg package reserved zero residual capacity, the residual bound was checked per plan only, and a plan's PnL booked an unhedged leg's cash as profit — so recorded paper cycles ended holding up to 0.006–0.007 BTC naked against a 0.002 limit, and the headline paper PnL across 44 cycles read **+698.91 USDT** while the marked equity change was **-25.46 USDT**. **(2) Spot inventory:** a spot sell was admitted against the position limit only, so a venue holding no BTC could sell (on the synthetic stream BYBIT sold **0.080 BTC it never held**). Both are fixed, tested and enabled in the platform configs.

2. Problems found (code before the change)

#	Where	Behaviour	Consequence
1	`risk.py` reserve(): `planned_residual = abs(sum(positions))`	BUY +q / SELL −q ⇒ 0 reserved	Any number of packages could leg out simultaneously
2	`oms.py` `_residual_within_bound`	`abs(plan.residual) < cap` per plan	Residuals of 0.001 each 'within bound' piled up (0.003–0.007 open)
3	`oms.py` `residual_risk_snapshot`	`open_residual_base` counted only non-CLOSED plans	Residual left by closed plans invisible to monitoring (no CRITICAL alert)
4	`oms.py` `_recompute_plan_economics`	`net_pnl = Σ sell cash − Σ buy cash − fees`	Orphan leg booked as ±78 USDT per 0.001 BTC
5	`risk.py` reserve(): `cash = {{buy_venue: notional}}`	Sell leg checked only vs `max_abs_position_base`	Venue with 0 BTC could sell (implicit short)
6	`risk.py` `AccountRiskSnapshot`	Static `available_cash` (usually ∞); `update_account_snapshot` never called	Risk never saw balances move with fills

3. Changes

3.1 Residual exposure (commit 40b52d4)

```
Admission (risk.py):
  planned_residual = max(|leg quantity|)                        # worst case, not the flat net
  refuse if |carried| ≥ max_residual_base                         → "carried residual at limit"
  refuse if reserved_residual + planned_residual + |carried|
            > max_legging_exposure_base (default max_position_base) → "max legging exposure exceeded"

Residual bound (oms.py):
  carried = Σ residual_base of CLOSED plans                       # derived from plans ⇒ restart-safe
  within  = residual == 0
            or ( |residual| < cap and ( |carried + residual| < cap or |carried + residual| ≤ |carried| ) )
  not within ⇒ HEDGE_UNWIND → unwind order, instead of closing with a naked leg

Marked PnL (oms.py, models.py):
  residual_mark_price = mean mid of the instrument's valid books (fallback: signal prices)
  net_pnl = gross_pnl + residual_base × residual_mark_price − fees − unwind_cost

Visibility: snapshot carried_residual_base, total_residual_base; monitoring CRITICAL on total_residual_base;
            analyze-session WARN "no_carried_residual"; config risk.max_legging_exposure_base.
```

3.2 Spot inventory (commits fa70b81, 8d66ce7)

```
risk.inventory.model: unconstrained (legacy default)  |  pre_funded
held_base(v)  = initial_base_by_venue[v]  + ledger.positions[(v, instrument)]
held_quote(v) = initial_quote_by_venue[v] + ledger.cash_deltas[v]       # quote default: initial_capital_by_venue
pre_funded, non-margin venues:
  BUY leg : reserved_quote(buy_venue) + buy_notional ≤ held_quote(buy_venue)  else "insufficient quote inventory: V"
  SELL leg: reserved_base(sell_venue) + quantity     ≤ held_base(sell_venue)  else "insufficient base inventory: V"
margin-enabled venue: existing margin checks, no base reserved
Reservation.base_by_venue released with the package; summaries report inventory model, label and balances.
Configs research.json / paper_watch.json: pre_funded, 0.05 BTC per venue + 10,000 USDT.
```

Commit	Subject	Files
`40b52d4`	Reserve worst-case legging exposure and bound residual in aggregate	docs/execution_realism.md | 29 +++++++++ src/crypto_stat_arb/config.py | 5 ++ src/crypto_stat_arb/models.py | 4 ++ src/crypto_stat_arb/monitoring.py | 6 ++ src/crypto_stat_arb/oms.py | 50 ++++++++++++++- src/crypto_stat_arb/risk.py | 27 +++++++- src/crypto_stat_arb/session_audit.py | 5 ++ tests/test_residual_exposure.py | 118 ++++++++++++++++++++++++++++++++++++ 8 files changed, 239 insertions(+), 5 deletions(-)
`fa70b81`	Add pre-funded spot inventory model to risk admission	docs/execution_realism.md | 32 +++++++++- src/crypto_stat_arb/config.py | 29 ++++++++- src/crypto_stat_arb/oms.py | 2 + src/crypto_stat_arb/risk.py | 96 ++++++++++++++++++++++++++++- src/crypto_stat_arb/runner.py | 3 + src/crypto_stat_arb/sdk.py | 3 +- src/crypto_stat_arb/session.py | 4 +- tests/test_spot_inventory.py | 138 ++ 8 files changed, 301 insertions(+), 6 deletions(-)

Commit	Subject	Files

`8d66ce7`	Use pre-funded inventory (0.05 BTC per venue) in research and paper_watch configs	`config/paper_watch.json | 7 +++++++`
		`config/research.json | 7 +++++++`
		`docs/execution_realism.md | 9 +++++++++`
		`tests/test_spot_inventory.py | 32 ++++++++++++++++++++++++++++++++`
		`tests/test_watch_pipeline.py | 18 ++++++++++++++++--`
		`5 files changed, 71 insertions(+), 2 deletions(-)`

4. Test results

Fresh run: **17/17 passed** (exit code 0); full suite: **250 passed, 3 warnings, 9 subtests passed in 13.14s**.

File	Class	Test	What it proves	Result	s
test_residual_exposure	WorstCaseReservationTests	`test_balanced_package_reserves_its_full_leg_quantity`	A flat BUY 0.01 / SELL 0.01 package reserves 0.01 (not 0). With a 0.015 legging limit the 2nd package is refused; after releasing the 1st it is admitted.	PASSED	0.001
test_residual_exposure	WorstCaseReservationTests	`test_carried_residual_counts_against_legging_exposure`	Carried residual −0.006 + new 0.01 > 0.015 limit → refused with 'max legging exposure exceeded'.	PASSED	0.000
test_residual_exposure	WorstCaseReservationTests	`test_new_risk_is_blocked_while_carried_residual_is_at_the_limit`	Carried residual 0.002 = max_residual_base 0.002 → all new risk refused ('carried residual at limit').	PASSED	0.000
test_residual_exposure	AggregateResidualBoundTests	`test_a_residual_that_offsets_the_carried_one_is_accepted`	Carried +0.01, cap 0.015: a further +0.01 is NOT within bound; a −0.01 that offsets it IS.	PASSED	0.002
test_residual_exposure	AggregateResidualBoundTests	`test_residuals_within_bound_one_at_a_time_cannot_pile_up`	Cap 0.02, Bybit leg always rejected: plan 1 closes holding 0.01; plan 2 would make 0.02 → it unwinds (UNWIND child) and closes flat. Carried and total residual stay 0.01.	PASSED	0.003
test_residual_exposure	MarkedPlanPnlTests	`test_hedged_plan_pnl_is_unchanged`	Fully hedged plan: residual 0, mark value 0, net_pnl = gross − fees (no change).	PASSED	0.001
test_residual_exposure	MarkedPlanPnlTests	`test_open_residual_is_marked_not_booked_as_cash`	Orphan buy 0.01 @ 60,000 (cash −600): residual marked at mid 60,030 → mark value 600.30, |net_pnl| < 1 instead of −600.	PASSED	0.001
test_spot_inventory	AdmissionTests	`test_buy_without_quote_on_the_buy_venue_is_rejected`	BINANCE holds 500 USDT; buying 0.01 × 60,000 = 600 → 'insufficient quote inventory: BINANCE'.	PASSED	0.000
test_spot_inventory	AdmissionTests	`test_margin_enabled_sell_venue_needs_no_base`	BYBIT margin-enabled with 0 BTC: sell admitted under margin checks, no base reserved.	PASSED	0.000
test_spot_inventory	AdmissionTests	`test_reserved_base_is_held_until_release`	BYBIT holds 0.015: 1st sell reserves 0.01, 2nd refused; release frees it; snapshot shows reserved 0.01 / available 0.005.	PASSED	0.000
test_spot_inventory	AdmissionTests	`test_sell_without_base_on_the_sell_venue_is_rejected`	BYBIT holds 0 BTC → selling 0.01 refused ('insufficient base inventory: BYBIT').	PASSED	0.000
test_spot_inventory	AdmissionTests	`test_unconstrained_model_keeps_the_legacy_behaviour`	No inventory policy: no base reserved; label UNCONSTRAINED_SPOT_SHORTING_ALLOWED.	PASSED	0.000
test_spot_inventory	LedgerDrivenBalanceTests	`test_fills_move_inventory_and_one_way_trading_runs_out_of_base`	Through the OMS + ledger: after one BUY BINANCE / SELL BYBIT package, BYBIT 0.01→0, BINANCE 0→0.01; the same direction is refused (no fills); the reverse trade is admitted and restores balances.	PASSED	0.003
test_spot_inventory	SharedInventoryTests	`test_one_directional_arb_drains_a_venue_shared_with_mean_reversion`	paper_watch config (pre-funded 0.05) on 6,000 synthetic events: base never negative, BINANCE+BYBIT = 0.10 BTC conserved, every refusal is 'insufficient base inventory: BYBIT'.	PASSED	0.669
test_spot_inventory	ConfigTests	`test_missing_section_is_unconstrained`	Config without risk.inventory → UNCONSTRAINED.	PASSED	0.001
test_spot_inventory	ConfigTests	`test_pre_funded_config_is_parsed_and_labelled`	pre_funded with 0.05 BTC/venue parsed; quote defaults to initial_capital_by_venue (10,000); label PRE_FUNDED_INVENTORY_NO_REBALANCING.	PASSED	0.001
test_spot_inventory	ConfigTests	`test_pre_funded_needs_a_base_balance_for_every_venue`	pre_funded missing a venue's base balance → config error.	PASSED	0.001

5. Data and results

5.1 Recorded live paper cycles (evidence of the problem)

44 cycles: headline `net_pnl` sum **698.91** USDT vs marked equity PnL **-25.46** USDT. 17 cycles ended holding a residual; the largest was 0.006 BTC (cap 0.002). Ten largest overstatements:

Cycle	Headline net_pnl	Equity PnL	Overstatement	End residual BTC
0010-20260914T162934Z	471.07	-1.62	472.69	-0.0060
0002-20260913T170854Z	-465.05	-1.44	-463.61	+0.0060
0011-20260914T170034Z	-396.32	-2.05	-394.27	+0.0050
0040-20260915T132143Z	339.01	-0.68	339.70	-0.0044
0008-20260914T152833Z	-250.13	-3.43	-246.69	+0.0031
0006-20260914T142731Z	231.74	-3.65	235.38	-0.0030
0029-20260915T052128Z	232.03	-0.38	232.41	-0.0030
0028-20260915T045053Z	231.29	-0.89	232.18	-0.0030
0003-20260913T173924Z	-155.19	-0.49	-154.69	+0.0020
0014-20260914T191733Z	78.96	-0.09	79.05	-0.0010

5.2 Replay of cycles 0006–0010: before vs after the residual fix

Cycle	Paper end res.	Before end res.	After end res.	net_pnl before	Equity before	net_pnl after	Equity after	Unwind fills	Within-bound closes	Opps	Kill switch
0006	-0.0030	+0.0000	**+0.0000**	-3.83	-3.83	**-4.70**	-4.64	0 → **8**	26 → 22	90 → 88	INACTIVE
0007	-0.0010	+0.0020	**+0.0000**	-160.39	-3.59	**-3.95**	-3.96	0 → **7**	26 → 12	72 → 75	INACTIVE
0008	+0.0031	+0.0020	**+0.0000**	-160.60	-3.47	**-4.31**	-4.34	0 → **10**	26 → 14	82 → 82	INACTIVE
0009	-0.0010	+0.0050	**-0.0010**	-397.20	-3.72	**-3.95**	-3.89	0 → **9**	31 → 25	81 → 82	INACTIVE
0010	-0.0060	-0.0070	**-0.0010**	549.79	-1.68	**-2.30**	-2.50	0 → **8**	17 → 7	43 → 45	INACTIVE

After: end residual ≤ 0.001 BTC on every cycle (cap 0.002); 7–10 residual unwinds per cycle where there were none; `net_pnl` within 0.2 USDT of equity PnL; kill switch INACTIVE, no open breaches, no open plans. The unwinds cost 0.3–0.8 USDT per cycle (spread paid to close positions that used to stay open).

5.3 Replay of cycles 0006–0010 with pre-funded inventory

Cycle	BTC/venue	Opps unconstrained → pre-funded	Refused	Equity unconstrained	Equity pre-funded	End BTC BINANCE	End BTC BYBIT
0006	0.05	88 → 88	0	-4.64	-4.64	0.05110	0.04890
0006	0.01	88 → 88	0	-4.64	-4.64	0.01110	0.00890
0007	0.05	75 → 75	0	-3.96	-3.96	0.04663	0.05337
0007	0.01	75 → 75	0	-3.96	-3.96	0.00663	0.01337
0008	0.05	82 → 82	0	-4.34	-4.34	0.048930	0.051070
0008	0.01	82 → 82	0	-4.34	-4.34	0.008930	0.011070
0009	0.05	82 → 82	0	-3.89	-3.89	0.04437000	0.054630
0009	0.01	82 → 82	0	-3.89	-3.89	0.00437000	0.014630
0010	0.05	45 → 45	0	-2.50	-2.50	0.056	0.043
0010	0.01	45 → 45	0	-2.50	-2.50	0.016	0.003

No package was refused at 0.05 or 0.01 BTC per venue; results are identical to unconstrained. The 0.001 BTC basis mean-reversion trades alternate direction, so base drifts ≤ 0.007 BTC per 30-minute cycle (closest: cycle 0010 BYBIT 0.003 at the 0.01 level).

5.4 Synthetic one-directional stream (6,000 events, paper_watch strategies)

Inventory	Plans (arb / MR)	Refusals	End BTC BINANCE	End BTC BYBIT	Ledger position BYBIT
pre-funded 0.05	180 / 24	insufficient base inventory: BYBIT: 199	0.100	0.000	-0.050
unconstrained	180 / 24	none	–	–	**-0.080** (sold without holding)

The synthetic arb is persistently BUY BINANCE / SELL BYBIT. Pre-funded: BYBIT's 0.05 BTC is drained after five 0.010 packages and every later package needing to sell there is refused — including the mean-reversion strategy's exits, which share the account. Unconstrained: BYBIT sells 0.080 BTC it never held.

5.5 Acceptance gate

Item	Value
Run	`local-all-20261012T151448Z`
Status	PASSED
Tests	250
Benchmark decision	PASS
Paper release id	`402f70d9-51ce-5b28-bcb4-a75ce4fe992d` (changed from bfbcab66…, expected after the config change)
`benchmark/backtest/summary.json`	PRE_FUNDED_INVENTORY_NO_REBALANCING · BTC {'BINANCE': '0.100', 'BYBIT': '0.000'}
`paper/summary.json`	PRE_FUNDED_INVENTORY_NO_REBALANCING · BTC {'BINANCE': '0.060', 'BYBIT': '0.040'}

6. Known gaps

- Quote reservation excludes fees; unwind (risk-reducing) orders are not blocked by inventory.
- A package's reservation is held until the plan closes, so between fill and close balances are counted twice (conservative).
- No inter-venue transfer/rebalancing model (results labelled `PRE_FUNDED_INVENTORY_NO_REBALANCING`); long one-directional runs exhaust a venue.

- Reservations are released at plan close, not reduced leg by leg as fills arrive.
- Two multi-strategy plumbing tests in `test_watch_pipeline.py` pin unconstrained inventory because the synthetic stream drains BYBIT.

Appendix A – all 44 recorded paper cycles

Cycle	Headline net_pnl	Equity PnL	End residual BTC
0001-20260913T170323Z	0.00	0.00	+0.0000
0001-20260914T113814Z	0.00	0.00	+0.0000
0002-20260913T170854Z	-465.05	-1.44	+0.0060
0002-20260914T115440Z	0.00	0.00	+0.0000
0003-20260913T173924Z	-155.19	-0.49	+0.0020
0003-20260914T123011Z	0.00	0.00	+0.0000
0004-20260913T215653Z	0.00	0.00	+0.0000
0004-20260914T131301Z	0.00	0.00	+0.0000
0005-20260914T040253Z	0.00	0.00	+0.0000
0005-20260914T135632Z	0.00	0.00	+0.0000
0006-20260914T063025Z	77.02	-1.27	-0.0010
0006-20260914T142731Z	231.74	-3.65	-0.0030
0007-20260914T145802Z	74.97	-3.43	-0.0010
0008-20260914T152833Z	-250.13	-3.43	+0.0031
0009-20260914T155903Z	74.91	-3.78	-0.0010
0010-20260914T162934Z	471.07	-1.62	-0.0060
0011-20260914T170034Z	-396.32	-2.05	+0.0050
0012-20260914T175138Z	0.00	0.00	+0.0000
0013-20260914T182555Z	0.00	0.00	+0.0000
0014-20260914T191733Z	78.96	-0.09	-0.0010
0015-20260914T201403Z	0.00	0.00	+0.0000
0016-20260914T205837Z	78.83	-0.10	-0.0010
0017-20260914T214147Z	0.00	0.00	+0.0000
0018-20260914T221421Z	0.00	0.00	+0.0000
0019-20260914T230108Z	0.00	0.00	+0.0000
0020-20260914T234336Z	0.00	0.00	+0.0000
0021-20260915T002852Z	0.00	0.00	+0.0000
0022-20260915T010026Z	0.00	0.00	+0.0000
0023-20260915T014745Z	0.00	0.00	+0.0000
0024-20260915T021917Z	0.00	0.00	+0.0000
0025-20260915T030354Z	77.38	-0.46	-0.0010
0026-20260915T034447Z	0.00	0.00	+0.0000
0027-20260915T041618Z	0.00	0.00	+0.0000
0028-20260915T045053Z	231.29	-0.89	-0.0030
0029-20260915T052128Z	232.03	-0.38	-0.0030
0030-20260915T060036Z	0.00	0.00	+0.0000
0031-20260915T063633Z	0.00	0.00	+0.0000
0032-20260915T071237Z	0.00	0.00	+0.0000
0033-20260915T075413Z	0.00	0.00	+0.0000
0034-20260915T085739Z	0.00	0.00	+0.0000
0035-20260915T095730Z	0.00	0.00	+0.0000
0036-20260915T103828Z	76.43	-0.70	-0.0010
0037-20260915T110840Z	-78.04	-0.99	+0.0010
0040-20260915T132143Z	339.01	-0.68	-0.0044

Appendix B – evidence bundle contents

File	Bytes	SHA-256
1_tests/junit_item1_item2.xml	2,680	8f4589358f8a21cc…
1_tests/pytest_full_suite.log	1,439	2f836986e85481c5…
1_tests/pytest_item1_item2.log	3,004	3673837bfd60a20c…
1_tests/test_cases.csv	1,874	4998ecb1b2332aef…
2_paper_residual/paper_cycles.csv	2,476	2fa6085c3fb1b612…
3_replay_residual/replay_before_after.csv	1,245	0d221f648065f664…
4_replay_inventory/inventory_levels.csv	1,446	e487ccb9e604bf56…
5_synthetic_drain/synthetic_drain.json	1,547	ee2c7402c9df1154…
6_acceptance/inventory_labels.json	329	d8e0d2c137e8c7ea…
6_acceptance/local_acceptance_summary.json	368	2768e6d8de194890…
7_code/40b52d4.patch	22,081	07cba9347b698864…
7_code/8d66ce7.patch	7,855	531d42cf2585f10a…
7_code/fa70b81.patch	23,540	097b00c895c5d263…
7_code/paper_watch.json	6,652	1e710d8de9a23de2…
7_code/research.json	5,509	5c6b27c0b10f2f7f…

```
Regenerate: python3.12 scripts/collect_inventory_residual_evidence.py --replay-dir <scratch>/div --staging-repo <clone>
```